

Simple SUITCASE-LIFT ALARM

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This suitcase-lift alarm circuit uses vibration sensor HDX-2NC instead of a commercially available sensor. A home-made mercury switch can also be used instead.

Circuit and working

Circuit diagram of the simple suitcase-lift alarm is shown in Fig. 1. It

is built around HDX-2NC, which is connected across connector CON2, transistors BC547 (T1) and BC557 (T2) and a piezo buzzer (PZ1).

Normally, vibration sensor terminals are shorted internally. When the suitcase is lifted, sensor terminals momentarily open. This allows capacitor C1 to charge through resistor R1 and diode D1. Discharging path is through VR1 and T1.

When T1 and T2 conduct due to charged C1, LED1 glows and the buzzer sounds. Alarm duration can be set using VR1.

Use a piezo buzzer with internal oscillator for the alarm sound. The circuit works off a 9V battery, which is connected across connector CON1.

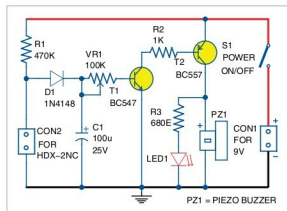


Fig. 1: Circuit diagram of the simple suitcase-lift alarm

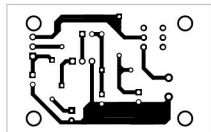


Fig. 2: An actual-size PCB pattern for the simple suitcase-lift alarm

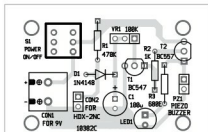


Fig. 3: Component layout of the PCB

PARTS LIST

Semiconductors:

- T1 - BC547 npn transistor
- T2 - BC557 pnp transistor
- D1 - 1N4148 signal diode
- LED1 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 470-kilo-ohm
- R2 - 1-kilo-ohm
- R3 - 680-ohm
- VR1 - 100-kilo-ohm potmeter

Capacitors:

- C1 - 100µF, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - 2-pin connector
- S1 - On/off switch
- PZ1 - Piezo buzzer
- 9V battery
- Vibration sensor HDX-2NC

Construction and testing

An actual-size, single-side PCB pattern for the simple suitcase-lift alarm is shown in Fig. 2 and its component layout in Fig. 3. After assembling the components on the PCB, house the circuit in a small box.

Check the circuit's working before you put it in your suitcase for use as an alarm. Switch on the circuit and place it in a suitable place inside your suitcase. **EZY**



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